

22/09/25

Ensemble Learning

$$\begin{aligned} \text{Total Error} &= E[(h_0(x) - y)^2] \\ &= E[(h_0(x) - \bar{h}(x))^2] + E[(\bar{h}(x) - \bar{y}(x))^2] + E[(\bar{y}(x) - y(x))^2] \\ &\quad \underbrace{\hspace{10em}}_{\text{variance}} \quad \underbrace{\hspace{10em}}_{\text{Bias}^2} \quad \underbrace{\hspace{10em}}_{\text{Noise}} \\ &\quad \text{Bagging} \quad \text{Boosting} \end{aligned}$$

Bagging (Bootstrap Aggregating)

Weak law of large number (WLLN)

random variable $x_i \leftarrow$ m no. of times but following iid

$$\frac{1}{m} \sum_{i=1}^m x_i \rightarrow \bar{x} \quad m \rightarrow \infty$$

$$\underline{h_0(x)} \rightarrow \underline{\bar{h}(x)} \quad \text{Var} = 0$$

$\mathcal{A}$ = Algo

Given m training data, $D_1, D_2, \dots, D_m \sim P^m$ $|D_i| = n$

$$D_i \leftarrow h_{D_i} \quad \bar{h} = \frac{1}{m} \sum_{i=1}^m h_{D_i} \rightarrow \bar{h} \quad m \rightarrow \infty$$

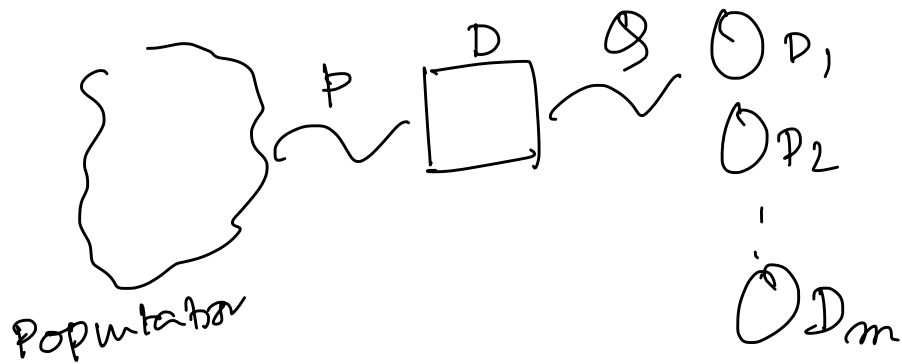
Bad news we do not have m data sets $D_1, D_2, \dots, D_m$
we have only D

Soln Bagging Simulate drawing from P by drawing uniformly with replacement from D

D
 $x_1 \ x_2 \ \dots \ x_n$
| | | | |
 $\frac{1}{n}$

$|D_i| = n$ D_m
 x_2
 x_5
 x_2
 x_1
 $\vdots$

$Q(x, y | D)$ = prob. distribution that I pick a training sample, (x, y) from D uniformly at random.



$$Q((x_i, y_i) | D) = \frac{1}{n} \quad \forall (x_i, y_i) \in D$$

$$|D| = n$$

$$\hat{h}_D = \frac{1}{m} \sum_{i=1}^m h_{D_i} \xrightarrow{m \rightarrow \infty} \bar{h}$$

(WLLN)
not iid

In practice,
bagging still reduces
variance.

$$\boxed{\text{Show: } Q = P}$$

lets assume that P is discrete.

$$P(X = x_i) = p_i$$

$$|D| = n$$

$$x_1, x_2, \dots, x_m$$

$Q(X = x_i)$ = what is the prob that x_i is sampled from the popu

$$= \sum_{k=1}^n \binom{n}{k} p_i^k (1-p_i)^{n-k} \quad \underbrace{\frac{k}{n}}_{\text{prob of picking one of the } k \text{ copies for } D \text{ to } D_i}$$

prob. that there are k copies of x_i in D

$$= \frac{1}{n} \sum_{k=1}^n \binom{n}{k} p_i^k (1-p_i)^{n-k} \cdot k$$

Expectation of Binomial distribution. with param $p_i = np_i$

$$= \frac{1}{n} n p_i = p_i$$

Steps

- Sample m data sets. $D_1, D_2, \dots, D_m$ with replacement from D
- For each D_j , train a classifier $h_j(\cdot)$
- The final classifier is $h(x) = \frac{1}{m} \sum_{j=1}^m h_j(x)$

m should not be very large as setting m uncwisly increase the time of the final classifier and it may result in diminishing return.



- Adv
- ① Easy to implement
 - ② Reduces variance
 - ③ Produces confidence intervals.
 - ④ Bagging is an unbiased estimator of the test error

Imagine one such D_i where (x_i, y_i) is not present

$$e_{\text{OOB}} = \frac{1}{n} \sum_{(x_i, y_i)} \ell(h_{D_i}(x_i), y_i)$$

out-of-bag error
OOB

Random Forest

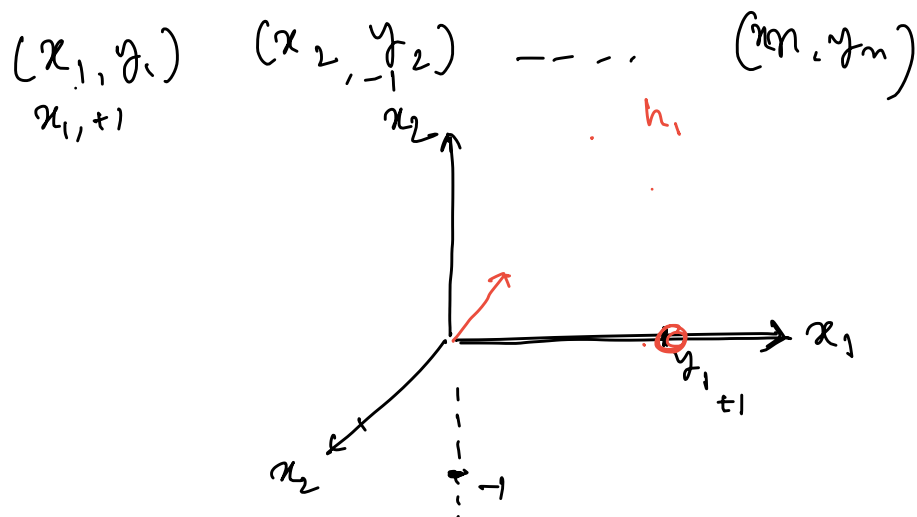
1. Sample m data sets $D_1, D_2, \dots, D_m$ with replacement from D
2. For each D_i , we will train a full DT $h_j(\cdot)$ with some modification:
before each split, we randomly sample K features from d features $K \leq d$ (with out replacement) and only consider them for the split

$$K = \sqrt{d}$$

Boosting

Can weak learners be combined to generate a strong learner with low bias?

create an ensemble classifier. $H_T(x) = \sum_{t=1}^T \alpha_t h_t(x)$



Boosting is a gradient descent on the loss function.

H_t : is the ensemble classifier that I have obtained so far at time t

$$h_{t+1} = \arg \min_{h \in \mathcal{H}} \underline{l(H_t + \alpha h)} \leftarrow$$

$$H_{t+1} = H_t + \alpha h_{t+1}$$

$$\langle h \cdot g \rangle$$

$$= \int_{\mathcal{X}} h(x) g(x) = \sum_{i=1}^n h(x_i) g(x_i)$$

$$f(H + \alpha h)$$

$$= f(H) + \alpha h f'(H) + \alpha^2 \frac{h^2}{2!} f''(H) + \dots$$

$$\approx f(H) + \alpha h \underline{f'(H)}$$

$$\checkmark \quad \underline{l(H + \alpha h)}$$

$$\approx \cancel{l(H)} + \alpha \langle \Delta l(H), h \rangle$$

$$\arg \min_{h \in \mathcal{H}} l(H + \alpha h) \approx \arg \min_{h \in \mathcal{H}} \langle \nabla l(H), h \rangle$$

$$= \arg \min_{h \in \mathcal{H}} \sum_{i=1}^n$$

